# HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

## PROFESSIONAL EXPERIENCE — 3.5 YEARS

**Software Engineer — HCL Technologies, Chennai, India** Sep 2021 – Jan 2025

- Developed and maintained **Python** data processing pipelines for analytics, automating ETL tasks and scheduled reporting across diverse data sources.
- Designed and implemented RESTful APIs using **Java Spring Boot** to facilitate cross-service communication and data exchange within enterprise systems.
- Managed production support for enterprise applications, responding to P1/P2 incidents and ensuring adherence to strict service level agreements.
- Contributed to the development of analytics and optimization software, implementing new modules and data-processing components as per project requirements.
- Performed analytical tasks involving **time-series data** and mathematical models to support optimization workflows within the energy technology domain.
- Researched and applied algorithms, data structures, and mathematical concepts to troubleshoot, debug, and enhance system performance.
- Containerised applications using **Docker** for consistent deployment across development and production environments on **AWS**.

## SKILLS

**Languages** – Python, Java, SQL, R

**Frameworks & Libraries** – Spring Boot, pandas, NumPy, scikit-learn, Flask, FastAPI

**Cloud & DevOps** – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git

**Databases** – PostgreSQL, MySQL, Oracle, NoSQL, Time-Series Databases

**Tools & Platforms** – Jira, Confluence, Splunk, Dynatrace, Linux

**Analytics & BI** – Power BI, Tableau, DAX, Power Query, Data Modelling

**Concepts** – APIs, Data Streams, Messaging Queues, Time-Series Data, Relational Data Modelling, Algorithms, Data Structures, Machine Learning Fundamentals

## PROJECTS

### Energy Analytics Platform Development

- Engineered **Python** modules for data ingestion and transformation, processing **time-series data** from energy sensors and external market feeds.
- Implemented REST APIs using **FastAPI** to serve processed data and model outputs to downstream analytics dashboards and reporting tools.
- Developed automated data validation scripts and unit tests to ensure data integrity and pipeline reliability, reducing data-related errors by approximately 18%.

### Time-Series Anomaly Detection System

- Developed a **Python**-based system to detect anomalies in sensor data streams using statistical methods and basic machine learning models.
- Applied **scikit-learn** for model training on historical time-series datasets, achieving a peak detection accuracy of ~82% on a validation set.
- Integrated the anomaly detection logic into a scheduled reporting workflow, generating alerts for critical deviations in system behaviour.

## EDUCATION

**M.Sc. Data Analytics — National College of Ireland, Dublin** Feb 2026

**B.E. Computer Science — SNS College of Technology, Coimbatore, India** 2021

## CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)